

SafeDrive: An Automatic Accident Management System

India is ranked second in the world for estimated road traffic deaths per year, according to WHO's global status report on road safety published in 2013. With such scary statistics, the need for an automatic accident management system is easy to sell

DILIN ANAND

SafeDrive is a personal emergency management device that automatically detects crashes or other crisis situations and makes that crucial first call for help. It also helps victims in less severe, yet highly-troubling, cases like tire blowout, vehicle breakdown and medical emergencies that happen on the road.

By connecting to the on-board diagnostics through any OBD-II port of the car, this device automatically monitors car operation in order to predict failures before these happen, at the

same time alerting users of best safe-driving practices. It was designed by a seven-member team led by Prasad Pillai, chief executive officer, and Dhanya CM, chief operating officer, Raksha SafeDrive.

How does the whole system work

SafeDrive works by employing the most-modern technologies in a very small package. In-built impact sensors detect accidents and collect high-accuracy location information from the on-board global positioning system (GPS). Then, this information is processed and transmitted by the SafeDrive device to central office via cellular technologies.

The SafeDrive ecosystem comprises two parts. The first part is a mobile Internet-enabled device. With voice capability similar to a phone, the SafeDrive device can automatically detect accidents and transmit distress messages with GPS location information.

The second part is a backend support network that mainly consists of 24×7-support personnel who can

receive distress messages and GPS location information transmitted by the SafeDrive device. Once the support centre receives the information, it is passed on to emergency providing agencies, the owner of the device is identified and emergency medical contacts are notified by SMSes and phone calls.

Hardware design

This device's hardware consists of three key sub-systems: connectivity, positioning and crash sensing.

Connectivity is achieved with a GPRS modem and associated circuitry. In India, cellular connectivity is the most ubiquitous and cost-effective networking option available. While it does not promise hundred per cent reach, it does provide the largest area of coverage that can be obtained compared to other technologies.

Positioning features like identifying the vehicle's location are very crucial to this product. Location sensing is provided by an in-built GPS circuitry.

The next subsystem is what designers claim to be the most crucial of all—the crash sensing circuitry. The circuitry has accelerometers to measure impact and sudden change in acceleration (deceleration) of the vehicle. Their proprietary algorithm then processes the digital data collected through these accelerometers and decides if the vehicle was involved in an accident.

Apart from these, SafeDrive also has supporting circuitry for receiving voice calls, in-built battery charging for uninterrupted power, data-logging

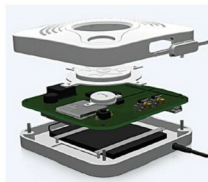


Fig. 1: Internal layout of the product

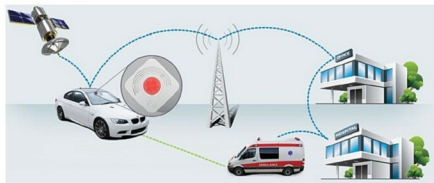


Fig. 2: The SafeDrive scenario